

DEVA ANAND

devaanand@umass.edu • (413) 315-1715 • devaanand.com • linkedin.com/in/devaaa • github.com/Deva-1903

Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Designed backend APIs for a multi-tenant NHS eTriage system (Node.js, MongoDB) serving **8+ UK clinical clients** and **10,000+ monthly submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack data-annotation and labeling platform (React, Node.js, MongoDB) used daily by the AI team to structure ML training data, cutting manual annotation by **100+ hours/month** and lifting throughput by **60%**.
- Owned backend reliability and performance: added integration tests, monitored production logs and debugged incidents, hardened auth and rate limiting with **AWS KMS** encryption, and remediated **20+ VAPT findings**.
- Served as primary technical POC, ran knowledge transfer onboarding a second developer in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot experiment for AI-assisted development.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM feature (**4-bit quantized Llama-7B via llama.cpp in C/C++**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with git-webhook auto-deploy, cron automation, and Stripe payments; live **3+ years**.

Projects

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication through cache-aware tiling, x86 **AVX/SIMD** vectorization, register-aware micro-kernels, and **OpenMP** multithreading, profiling L1/L3/TLB performance with **perf** and inspecting generated assembly.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a large-scale distributed Spark ETL pipeline over **~1.4B rows**, cutting end-to-end runtime **~25%** and the tail-latency ratio **2.14x to 1.74x** via broadcast joins, AQE, skew handling, and partitioning.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI backend service with typed retrieval planning, reranking, and deterministic extraction returning **cell-level provenance**; hardened with **150+ tests** and provider routing across backends.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: C++, Python, TypeScript, JavaScript, SQL, Java, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: FastAPI, Node.js, Express, React, Docker, Git, GitHub Actions, REST APIs

Systems / Performance: AVX/SIMD (x86), OpenMP multithreading, cache-aware design, Spark / PySpark, distributed data processing, perf profiling, Linux

ML & AI: PyTorch, Hugging Face Transformers, embedding search, RAG, LLM integration, NumPy, evaluation

Cloud / Platforms: AWS, GCP, Azure, Docker, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.